

Linear algebra – 6

Linear algebra in combinatorics

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Theoretical minimum.

Def. The *dimension method*: to bound the size of a family of objects, attach to each object a vector in a fixed space and prove that these vectors are linearly independent; then the size is at most the dimension. The characteristic vector of a set $S \subseteq [n]$ is $\mathbf{1}_S \in \mathbb{F}_2^n$ or $\in \mathbb{R}^n$, and $\langle \mathbf{1}_S, \mathbf{1}_T \rangle = |S \cap T|$.

Theorem 1 (Oddtown). If $A_1, \dots, A_m \subseteq [n]$ satisfy $|A_i|$ odd and $|A_i \cap A_j|$ even for $i \neq j$, then $m \leq n$.

Theorem 2 (Fisher 1940). If $A_1, \dots, A_m \subseteq [n]$ satisfy $|A_i \cap A_j| = \lambda$ for a fixed $\lambda \geq 1$ and all $i \neq j$, then $m \leq n$.

Fact 1. $\text{rank}(A + B) \leq \text{rank } A + \text{rank } B$ and a rank- r matrix is a sum of r matrices of rank 1; this is the whole content of the Graham–Pollak argument.

1. Prove the Oddtown theorem and show that it is sharp. Then prove the Eventown bound $m \leq 2^{\lfloor n/2 \rfloor}$ for families with all $|A_i|$ and all $|A_i \cap A_j|$ even, and explain why the two answers differ so wildly.
2. (IMC) Let n be a positive integer. At most how many distinct unit vectors can be chosen in $\mathbb{R}^n$ so that among any three of them at least two are orthogonal?
3. (IMC) Determine the smallest possible rank of an $n \times n$ matrix that has zeros on the main diagonal and strictly positive real numbers off the main diagonal.
4. (Putnam 2018) Let $S_1, \dots, S_{2^n-1}$ be the non-empty subsets of $\{1, \dots, n\}$ in some order and let M be the $(2^n - 1) \times (2^n - 1)$ matrix with entries

$m_{ij} = 0$ if $S_i \cap S_j = \emptyset$ and $m_{ij} = 1$ otherwise. Compute $\det M$ and check that it does not depend on the ordering.

5. (Mexmat, 2017) Let $x_1, \dots, x_n, y_1, \dots, y_n$ be pairwise distinct points of $\mathbb{R}^d$ with $d \leq 2n - 2$ such that $|x_i - y_i| = a$ for all i while all the remaining distances between distinct points of the collection are equal to the same number b . Prove that all these points lie in an n -dimensional affine subspace of $\mathbb{R}^d$.